

## NEWHAVEN LH, United Kingdom

WMO: 038800

Lat: 50.7818N

Lon: 0.0571E

Elev: 16

StdP: 14.69

Time Zone: 0.00 (EUW)

Period: 90-00

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 28.7       | 31.4 | 23.6                          | 17.8 | 30.6 | 26.3 | 20.3 | 33.3 | 42.4                  | 48.2 | 38.1 | 48.2 | 10.5                  | 350  | 0.663 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 8             | 8.8                       | 75.1            | 64.9 | 71.8 | 64.0 | 69.1 | 62.6 | 66.6                | 72.3 | 65.1 | 69.7 | 63.8 | 67.8 | 12.0                    | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 64.4                            | 90.7 | 69.1 | 63.1 | 86.5 | 67.5 | 62.0 | 83.0 | 66.3 | 31.1          | 72.9 | 30.0 | 70.2 | 29.1 | 67.9 | 73.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 33.5          | 30.0 | 26.4 | 25.6                       | 80.3 | 4.0                | 3.1 | 22.7                                               | 82.5 | 20.3       | 84.3 | 18.1       | 86.1 | 15.2       | 88.3 |    |    |
| (5)               |      |      | 24.0                       | 69.1 | 4.0                | 2.7 | 21.2                                               | 71.0 | 18.8       | 72.6 | 16.6       | 74.1 | 13.7       | 76.0 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 51.9   | 43.1  | 42.6  | 45.5  | 47.8  | 54.1  | 57.9  | 62.3  | 63.7  | 59.3  | 54.0  | 48.1  | 43.7  |  |  |
|                                                                        | DBStd       | 8.65   | 5.24  | 5.34  | 3.51  | 4.30  | 4.98  | 3.39  | 3.38  | 3.86  | 3.62  | 5.04  | 5.70  | 5.51  |  |  |
|                                                                        | HDD50       | 999    | 214   | 208   | 142   | 89    | 15    | 1     | 0     | 0     | 0     | 23    | 103   | 204   |  |  |
|                                                                        | HDD65       | 4837   | 678   | 627   | 604   | 515   | 339   | 217   | 100   | 71    | 174   | 342   | 508   | 662   |  |  |
|                                                                        | CDD50       | 1688   | 1     | 0     | 3     | 24    | 142   | 237   | 380   | 423   | 279   | 146   | 45    | 8     |  |  |
|                                                                        | CDD65       | 53     | 0     | 0     | 0     | 0     | 1     | 3     | 16    | 30    | 3     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 140    | 0     | 0     | 0     | 0     | 2     | 8     | 43    | 82    | 5     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 11     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 10    | 0     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 12.4   | 14.8  | 13.1  | 11.9  | 12.0  | 11.1  | 11.3  | 11.2  | 10.8  | 11.7  | 13.5  | 13.2  | 14.2  |  |  |
| (15) Precipitation                                                     | PrecAvg     | 32.3   | 3.7   | 2.5   | 1.9   | 2.0   | 1.9   | 2.1   | 2.1   | 2.4   | 2.4   | 3.7   | 4.0   | 3.6   |  |  |
|                                                                        | PrecMax     | 47.0   | 7.4   | 6.1   | 5.2   | 4.3   | 4.3   | 6.5   | 4.8   | 5.2   | 5.1   | 11.1  | 9.1   | 6.8   |  |  |
|                                                                        | PrecMin     | 24.1   | 0.7   | 0.3   | 0.1   | 0.2   | 0.4   | 0.4   | 0.4   | 0.1   | 0.2   | 0.7   | 1.3   | 0.5   |  |  |
|                                                                        | PrecStd     | 5.5    | 1.7   | 1.5   | 1.0   | 1.3   | 1.1   | 1.5   | 1.1   | 1.3   | 1.5   | 2.1   | 2.1   | 1.9   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 52.3  | 52.2  | 56.0  | 63.0  | 73.4  | 74.9  | 78.6  | 81.5  | 72.9  | 64.9  | 59.6  | 55.6  |  |  |
|                                                                        |             | MCWB   | 50.3  | 49.8  | 50.8  | 54.9  | 61.7  | 63.8  | 66.8  | 67.1  | 63.4  | 61.5  | 57.1  | 53.1  |  |  |
|                                                                        | 2%          | DB     | 51.4  | 50.9  | 53.0  | 58.7  | 68.2  | 68.7  | 73.9  | 75.8  | 68.1  | 63.4  | 58.3  | 54.4  |  |  |
|                                                                        |             | MCWB   | 49.6  | 48.8  | 48.8  | 52.5  | 59.3  | 60.7  | 65.1  | 65.7  | 62.0  | 60.2  | 56.2  | 52.6  |  |  |
|                                                                        | 5%          | DB     | 50.7  | 49.8  | 51.2  | 55.7  | 64.7  | 65.8  | 70.4  | 72.5  | 66.3  | 62.2  | 57.1  | 53.0  |  |  |
|                                                                        |             | MCWB   | 49.1  | 47.7  | 48.3  | 50.8  | 58.2  | 59.3  | 63.0  | 64.8  | 61.1  | 59.3  | 55.0  | 51.4  |  |  |
|                                                                        | 10%         | DB     | 49.9  | 48.8  | 50.1  | 53.3  | 61.8  | 63.3  | 67.7  | 69.8  | 64.9  | 61.0  | 55.9  | 51.6  |  |  |
|                                                                        |             | MCWB   | 48.5  | 47.0  | 47.8  | 49.9  | 56.5  | 58.4  | 61.7  | 63.8  | 60.3  | 58.1  | 53.7  | 49.9  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 51.1  | 50.6  | 51.8  | 55.9  | 63.4  | 64.5  | 69.2  | 69.3  | 65.3  | 62.4  | 57.6  | 53.9  |  |  |
|                                                                        |             | MCDB   | 51.6  | 51.6  | 54.3  | 61.2  | 70.3  | 71.8  | 76.1  | 76.6  | 70.2  | 64.0  | 58.8  | 54.7  |  |  |
|                                                                        | 2%          | WB     | 50.2  | 49.3  | 50.1  | 53.2  | 60.6  | 61.9  | 66.2  | 67.0  | 63.7  | 61.1  | 56.8  | 53.0  |  |  |
|                                                                        |             | MCDB   | 51.0  | 50.5  | 51.6  | 57.5  | 66.6  | 66.7  | 71.9  | 72.7  | 67.3  | 62.8  | 58.0  | 54.1  |  |  |
|                                                                        | 5%          | WB     | 49.5  | 48.3  | 49.3  | 51.5  | 58.6  | 60.2  | 64.1  | 65.7  | 62.3  | 59.9  | 55.5  | 51.8  |  |  |
|                                                                        |             | MCDB   | 50.4  | 49.4  | 50.6  | 54.2  | 63.9  | 64.2  | 68.4  | 70.7  | 65.4  | 61.8  | 56.8  | 53.0  |  |  |
|                                                                        | 10%         | WB     | 48.7  | 47.4  | 48.5  | 50.5  | 56.7  | 59.0  | 62.7  | 64.6  | 61.2  | 58.7  | 54.1  | 50.1  |  |  |
|                                                                        |             | MCDB   | 49.7  | 48.5  | 49.8  | 52.6  | 61.1  | 62.2  | 65.9  | 68.7  | 64.1  | 60.8  | 55.5  | 51.4  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 6.1   | 6.3   | 6.8   | 8.0   | 9.0   | 7.5   | 7.7   | 8.8   | 8.6   | 7.9   | 7.0   | 6.4   |  |  |
|                                                                        |             | MCDBR  | 5.4   | 5.6   | 7.6   | 11.7  | 14.2  | 12.6  | 13.5  | 12.5  | 9.9   | 7.2   | 5.7   | 5.7   |  |  |
|                                                                        |             | MCWBR  | 5.6   | 5.3   | 5.8   | 7.4   | 8.9   | 7.4   | 7.5   | 6.7   | 6.6   | 6.2   | 5.3   | 6.3   |  |  |
|                                                                        | 5% WB       | MCDBR  | 5.4   | 5.2   | 6.0   | 8.9   | 13.4  | 10.4  | 10.7  | 10.0  | 8.2   | 6.8   | 4.8   | 6.1   |  |  |
|                                                                        |             | MCWBR  | 5.7   | 5.5   | 5.1   | 6.3   | 8.7   | 6.6   | 7.0   | 6.4   | 6.9   | 6.6   | 4.9   | 6.8   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.335 | 0.352 | 0.391 | 0.408 | 0.414 | 0.413 | 0.407 | 0.401 | 0.384 | 0.368 | 0.340 | 0.326 |  |  |
|                                                                        | taud        |        | 2.363 | 2.364 | 2.257 | 2.237 | 2.278 | 2.307 | 2.330 | 2.367 | 2.399 | 2.415 | 2.408 | 2.357 |  |  |
|                                                                        | Ebn at Noon |        | 215   | 245   | 254   | 264   | 267   | 268   | 263   | 263   | 255   | 236   | 213   | 199   |  |  |
|                                                                        | Edh at Noon |        | 21    | 27    | 35    | 40    | 40    | 39    | 38    | 35    | 30    | 25    | 20    | 18    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 263   | 489   | 871   | 1394  | 1666  | 1846  | 1762  | 1472  | 1092  | 617   | 312   | 203   |  |  |
|                                                                        | RadStd      |        | 21    | 64    | 107   | 123   | 108   | 160   | 140   | 105   | 75    | 54    | 34    | 24    |  |  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (69 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page